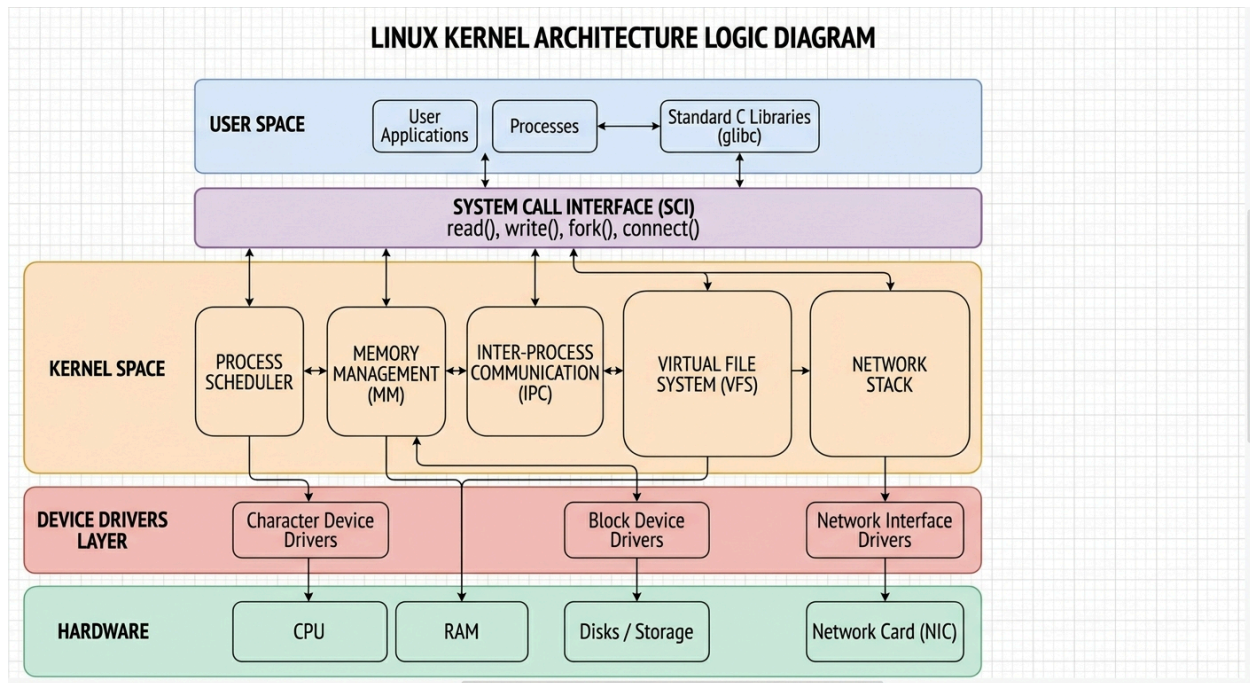


Name - Saikat Sheet
Batch - 25SUB5936_Linux System Programming
ID - 68500

Task 1:

- Kernel Architecture Diagram - Draw a detailed diagram of the Linux kernel architecture.
- Label and write a short description (2-3 sentences) for each major component like Scheduler, File System, Network Stack, etc.



Here is a refined description of the major subsystems of the Linux Kernel, focusing on their core logical functions:

1. Process Scheduler

The Scheduler is responsible for managing CPU time by deciding which active process runs at any given moment. It ensures fair distribution of processing power and maintains system responsiveness by switching between tasks based on priority and state.

2. Memory Management (MM)

This component handles the allocation and deallocation of physical and virtual memory for all running processes. It ensures that processes are isolated from one another and manages "swapping" to disk when the system runs out of physical RAM.

3. Virtual File System (VFS)

The VFS provides a unified interface that allows users to interact with different types of storage and file systems (like Ext4, NTFS, or NFS) using the same commands. It abstracts the underlying hardware details so that applications can read and write data without knowing the specific format of the disk.

4. Network Stack

The Network Stack implements various communication protocols, primarily the TCP/IP suite, to manage data movement across networks. It handles everything from routing and packet assembly to ensuring data reaches the correct application via specific ports.

5. Inter-Process Communication (IPC)

The IPC subsystem provides the mechanisms that allow different processes to talk to each other and synchronize their actions. It manages shared memory, signals, and message queues so that complex applications can function as a coordinated unit.

6. Device Drivers

Drivers are specialized modules that act as the bridge between the kernel's logical subsystems and the physical hardware. They translate generic OS requests into the specific electronic signals required to operate devices like hard drives, keyboards, and network cards.
